Software Requirements Specification (SRS)

Project: Online Exam Portal
Team Number: 9
Version: 1.1
Authors: Aadish Sarin (Lead), A S Vincent, Aditya Rao, Akshansh Agrawal
Date: 09-09-2026
Status: Final

Revision history

Version	Date	Author	Change summary	Approval
1.0	06-09-2026	Aadish Sarin	Initial SRS draft for Online Exam Portal	
1.1	09-09-2026	Aadish Sarin	Finalized SRS; completed UML diagrams and RTM	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
Team Lead	Aadish Sarin (PES2UG24CS007)		

Team & Work Allocation

Work is divided equally across the four members. Each member owns one core module (4 functional requirements each), one shared SRS chapter, and a share of the NFR, security, RTM and use-case work. Aadish Sarin, as team lead, additionally coordinates and compiles the final document.

SRN	Member	Core Module	FR IDs	Shared Chapter
PES2UG24CS002	A S Vincent	M1 - Authentication & User/Role Management	OEP-F-001..004	Ch.2 Overall Description
PES2UG24CS007	Aadish Sarin (Lead)	M2 - Exam & Question-Bank Management	OEP-F-005..008	Ch.1 Introduction + Integration
PES2UG24CS034	Aditya Rao	M3 - Exam Delivery / Test-Taking Engine	OEP-F-009..012	Ch.3 External Interfaces
PES2UG24CS045	Akshansh Agrawal	M4 - Evaluation, Results & Analytics	OEP-F-013..016	Ch.6 Quality Attributes

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1. Introduction

Owner: Aadish Sarin (PES2UG24CS007)

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Online Exam Portal. It defines the functional and non-functional requirements, interfaces, and verification criteria for a web-based platform that lets instructors create and schedule exams, students attempt them online, and the system evaluate and report results.

1.2 Scope

Covers user registration and authentication, role-based access (Student, Instructor, Admin), exam and question-bank management, timed online exam delivery, automatic and manual evaluation, result publication and analytics. Excludes payment/billing, proctoring hardware, and integration with external LMS beyond documented APIs.

1.3 Audience

Developers, QA Engineers, Instructors, Students, System Administrators, and Assessment Evaluators.

1.4 Definitions

List of acronyms: OEP (Online Exam Portal), MCQ, RBAC, OTP, UI, API, TLS, RTM, WCAG.

2. Overall description

Owner: A S Vincent (PES2UG24CS002)

2.1 Product perspective

The Online Exam Portal is a web application with a browser-based front end, an application/API server, and a relational database. It interacts with users over HTTPS and optionally with an email/SMS gateway for notifications and OTP.

2.2 Major product functions (detailed)

- Register and authenticate users
- Manage roles (Student / Instructor / Admin)
- Create, edit and schedule exams
- Maintain a question bank (MCQ, True/False, short answer)
- Deliver timed exams with auto-save and auto-submit
- Auto-grade objective questions and support manual grading
- Publish results and generate analytics/reports

2.3 User roles and characteristics (expanded)

- Student: attempts exams, views results; expects a simple, reliable interface.
- Instructor: creates exams/questions, grades subjective answers, views analytics.
- Admin: manages users, roles, and system configuration.

2.4 Operating environment

Modern web browsers (Chrome, Edge, Firefox), server on Linux with an application runtime and a relational database (e.g., MySQL/PostgreSQL), hosted on-premise or cloud.

2.5 Constraints

Responsive web UI, secure handling of credentials and exam content, reasonable performance under concurrent exam load, and data privacy of student records.

3. External interface requirements

Owner: Aditya Rao (PES2UG24CS034)

3.1 User interfaces

Responsive web UI with role-specific dashboards (Student, Instructor, Admin). Exam screen shows a countdown timer, question navigation panel, and mark-for-review. Accessibility: keyboard navigation and high-contrast mode.

3.2 Hardware interfaces

- Standard client devices (desktop/laptop/tablet) with a web browser
- Server hardware / cloud VM hosting the application and database

3.3 Software interfaces

- REST API (JSON over HTTPS) between front end and application server
- Database interface for persistence of users, exams, questions, submissions and results
- Email/SMS gateway API for notifications and OTP

3.4 Communications

- HTTPS/TLS 1.2+ enforced for all client-server traffic
- Periodic auto-save of answers and safe re-connection handling during an exam

4. System features (detailed)

Each requirement includes acceptance criteria and a reference test case. IDs follow OEP-F-###.

4.1 Authentication & User/Role Management (Module M1)

Owner: A S Vincent (PES2UG24CS002)

Description: Register and authenticate users and enforce role-based access for Student, Instructor and Admin.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OEP-F-001	The system shall allow a new user to register with name, email, password and role request.	Functional	High	Student/Admin	AC-F-001: Valid details create an account pending role assignment. Test: TC-Auth-01	Email uniqueness check
OEP-F-002	The system shall authenticate users via email and password and create a secure session.	Functional	High	Security	AC-F-002: Correct credentials grant a role-based session. Test: TC-Auth-02	Session/token management
OEP-F-003	The system shall allow password reset via an OTP/link sent to the registered email.	Functional	Medium	Student	AC-F-003: Valid OTP allows password change. Test: TC-Auth-03	Email/SMS gateway
OEP-F-004	The system shall enforce role-based access control (Student/Instructor/Admin) on all features.	Functional	High	Security	AC-F-004: Users can access only permitted	RBAC middleware

					pages/APIs. Test: TC-Auth-04	

4.2 Exam & Question-Bank Management (Module M2)

Owner: Aadish Sarin (PES2UG24CS007)

Description: Instructors create and manage exams and maintain a reusable question bank.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OEP-F-005	The system shall allow an instructor to create, edit and delete an exam.	Functional	High	Instructor	AC-F-005: CRUD operations on exams persist correctly. Test: TC-Exam-01	Instructor role required
OEP-F-006	The system shall allow adding questions (MCQ, True/False, short answer) to a question bank.	Functional	High	Instructor	AC-F-006: Questions saved with type, options and correct answer. Test: TC-Exam-02	Question schema
OEP-F-007	The system shall let the instructor set exam schedule, duration and total marks.	Functional	High	Instructor	AC-F-007: Exam has start/end time, duration and marks. Test: TC-Exam-03	Server clock/timezone
OEP-F-008	The system shall shuffle/randomize question order per student when	Functional	Medium	Instructor	AC-F-008: Two students see questions in	Depends on OEP-F-006

	enabled.				different order. Test: TC-Exam-04	

4.3 Exam Delivery / Test-Taking Engine (Module M3)

Owner: Aditya Rao (PES2UG24CS034)

Description: Deliver timed exams to students with reliable saving and submission.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OEP-F-009	The system shall show a student the list of available/scheduled exams.	Functional	High	Student	AC-F-009: Only eligible, in-window exams are listed. Test: TC-Take-01	Depends on OEP-F-007
OEP-F-010	The system shall present the exam with a visible countdown timer.	Functional	High	Student	AC-F-010: Timer counts down and is enforced server-side. Test: TC-Take-02	Server-side time enforcement
OEP-F-011	The system shall auto-save answers periodically and auto-submit on timeout.	Functional	High	Student	AC-F-011: Answers persist on refresh; exam submits at time-up. Test: TC-Take-03	Auto-save interval config
OEP-F-012	The system shall allow navigation between questions and mark-for-	Functional	Medium	Student	AC-F-012: Student can jump to any question	UI navigation panel

	review.				and flag it. Test: TC-Take-04	

4.4 Evaluation, Results & Analytics (Module M4)

Owner: Akshansh Agrawal (PES2UG24CS045)

Description: Grade submissions, publish results and provide analytics.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OEP-F-013	The system shall auto-grade objective questions (MCQ/True-False) on submission.	Functional	High	Instructor	AC-F-013: Score computed correctly against answer key. Test: TC-Eval-01	Depends on OEP-F-011
OEP-F-014	The system shall provide a manual grading interface for subjective answers.	Functional	High	Instructor	AC-F-014: Instructor assigns marks; total updates. Test: TC-Eval-02	Instructor role
OEP-F-015	The system shall publish results and let a student view their scorecard.	Functional	High	Student	AC-F-015: Published result visible only to the student. Test: TC-Eval-03	RBAC (OEP-F-004)
OEP-F-016	The system shall generate analytics/reports (average, pass %, item analysis) for instructors.	Functional	Medium	Instructor	AC-F-016: Report values match underlying data. Test: TC-Eval-04	Depends on OEP-F-013

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs OEP-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement	Owner
OEP-NF-001	Exam pages and question navigation shall load in <= 3 seconds for 90% of requests under normal load.	Performance	High	90th percentile <= 3s in load test. Test: TC-Perf-01	Aditya Rao
OEP-NF-002	The system shall support at least 500 concurrent students taking exams without failure.	Scalability	High	Load test with 500 virtual users passes. Test: TC-Perf-02	Aadish Sarin
OEP-NF-003	The system shall provide 99.5% availability during scheduled exam windows and recover in-progress answers after a client disconnect.	Reliability	High	Uptime report >= 99.5%; answers restored after reconnect. Test: TC-Rel-01	Akshansh Agrawal
OEP-NF-004	All credentials shall be hashed and all data transmitted over TLS 1.2+.	Security/Compliance	High	No plaintext passwords; TLS enforced. Test: TC-Sec-01	A S Vincent
OEP-NF-005	The UI shall be responsive and conform to WCAG 2.1 AA where	Usability/Accessibility	Medium	Accessibility and responsiveness audit pass. Test: TC-UX-01	Akshansh Agrawal

	applicable.				

5.1 Security

5.1.1 Security Objectives

SO-1 (Owner: Aadish Sarin): Ensure confidentiality and integrity of exam content and results so that only authorized users can access exams, answers and marks.

SO-2 (Owner: Aditya Rao): Prevent malpractice and tampering during exams and maintain a trustworthy audit trail of submissions and grading.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref	Owner
OEP-SR-001	TLS 1.2+ shall be mandatory for all network connections.	Security	High	All traffic over HTTPS; HTTP redirected. Test: TC-Sec-02	Aadish Sarin
OEP-SR-002	Passwords shall be stored using a salted strong hash (bcrypt/argon2).	Security	High	No reversible passwords in DB. Test: TC-Sec-03	A S Vincent
OEP-SR-003	RBAC shall be enforced server-side on every protected API endpoint.	Security	High	Unauthorized role receives 403. Test: TC-Sec-04	A S Vincent
OEP-SR-004	Exam questions shall not be exposed before the scheduled start and correct answers shall never be sent to the client during an exam.	Security	High	No answers in client payload; questions locked pre-start. Test: TC-Sec-05	Aditya Rao
OEP-SR-005	The system shall keep an audit log of login, submission and grading actions.	Security	Medium	Audit records created and retained. Test: TC-Sec-06	Akshansh Agrawal

6. Quality attributes & Acceptance tests

Owner: Akshansh Agrawal (PES2UG24CS045)

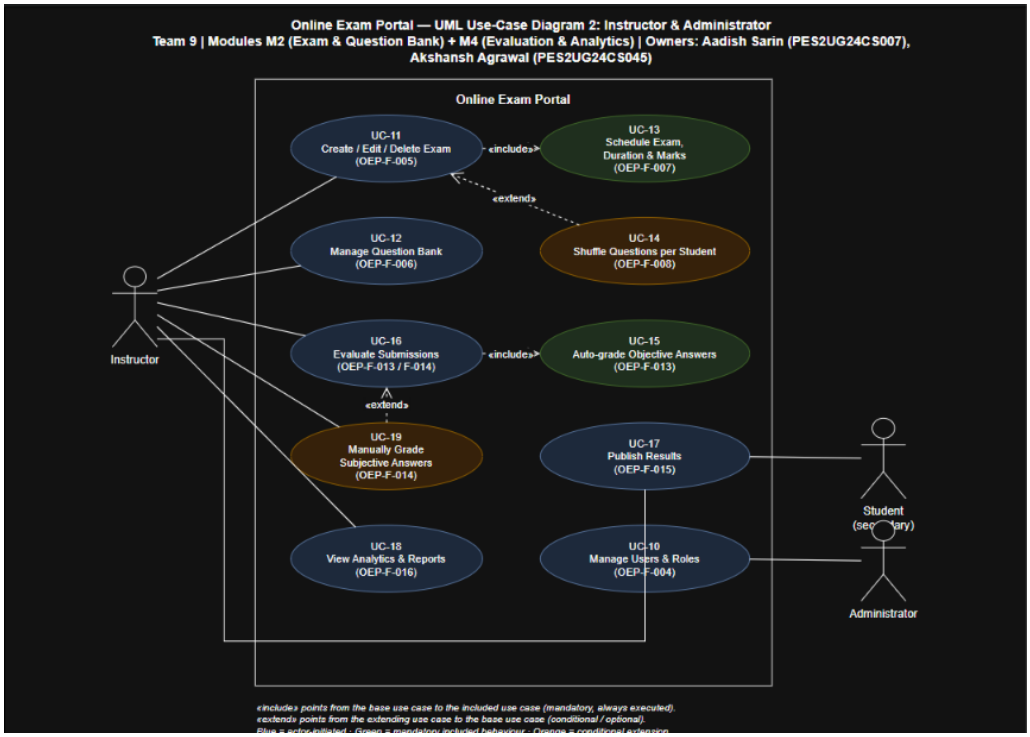
- Exit criteria for acceptance: all High-priority functional requirements implemented and verified, no critical NFR failures, and the RTM shows all test cases passed.

- Acceptance test suites: Authentication, Exam Management, Exam Delivery, Evaluation, Performance, Security, and Accessibility tests.

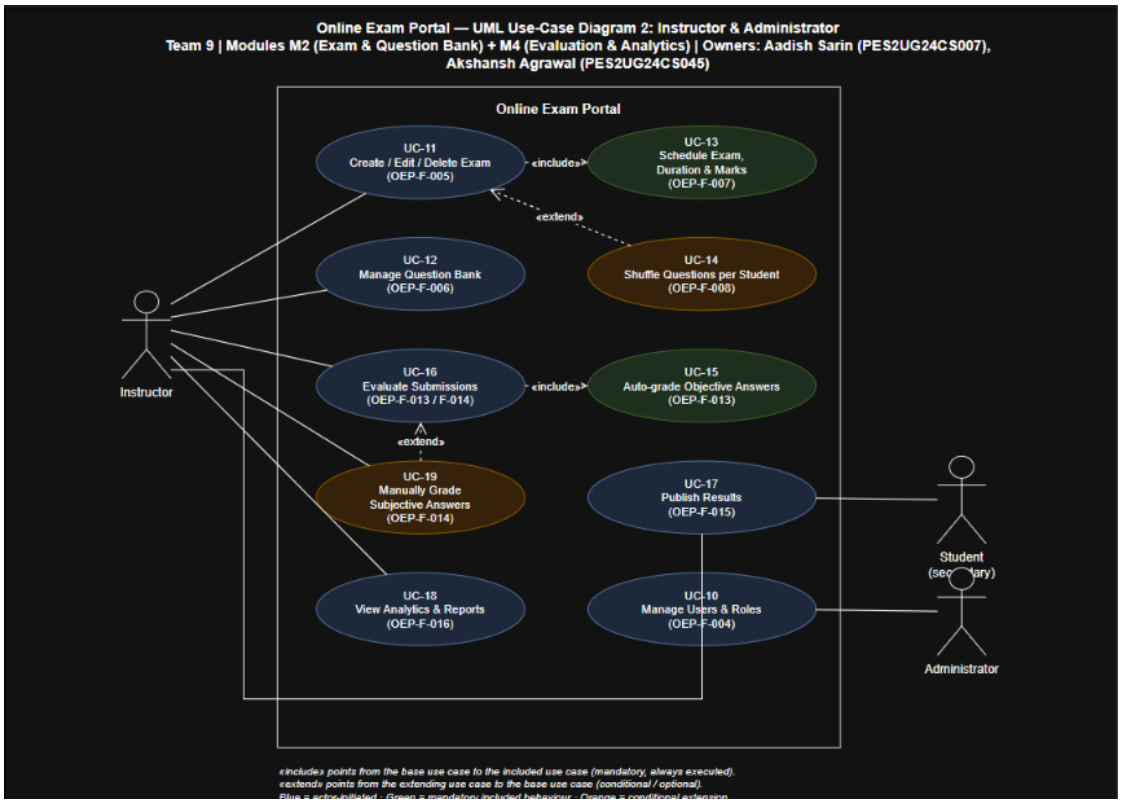
7. UML Use-Case Diagrams

7.1 Use-Case Diagrams

The following UML use-case diagrams model the primary interactions of the Online Exam Portal: (a) Student flow: registration/login, viewing available exams, taking an exam, and viewing results (Owner: A S Vincent + Aditya Rao).



(b) Instructor/Admin flow: user management, exam and question management, grading, and analytics. (Owner: Aadish Sarin + Akshansh Agrawal).



8. Requirements Traceability Matrix (RTM)

Owner: Aadish Sarin (PES2UG24CS007) - compiles from all modules

Req ID	Requirement short	SRS Section / Module Reference	Module	Test case(s)	Status (N/P/A)	Comments
OEP-F-001	Register user	4.1 / Authentication & User/Role Management	AuthModule	TC-Auth-01	N	
OEP-F-002	Login/session	4.1 / Authentication & User/Role Management	AuthModule	TC-Auth-02	N	
OEP-F-003	Password reset	4.1 / Authentication & User/Role Management	AuthModule	TC-Auth-03	N	
OEP-F-004	RBAC	4.1 / Authentication & User/Role Management	AuthModule	TC-Auth-04	N	
OEP-F-005	Create/edit/delete exam	4.2 / Exam & Question-Bank Management	ExamModule	TC-Exam-01	N	
OEP-F-006	Question bank	4.2 / Exam & Question-Bank Management	ExamModule	TC-Exam-02	N	
OEP-F-007	Exam schedule/duration/marks	4.2 / Exam & Question-Bank Management	ExamModule	TC-Exam-03	N	

OEP-F-008	Shuffle questions	4.2 / Exam & Question-Bank Management	ExamModule	TC-Exam-04	N	
OEP-F-009	Available exams	4.3 / Exam Delivery / Test-Taking Engine	DeliveryModule	TC-Take-01	N	
OEP-F-010	Timed delivery	4.3 / Exam Delivery / Test-Taking Engine	DeliveryModule	TC-Take-02	N	
OEP-F-011	Auto-save/submit	4.3 / Exam Delivery / Test-Taking Engine	DeliveryModule	TC-Take-03	N	
OEP-F-012	Question navigation/mark review	4.3 / Exam Delivery / Test-Taking Engine	DeliveryModule	TC-Take-04	N	
OEP-F-013	Auto-grade	4.4 / Evaluation, Results & Analytics	EvalModule	TC-Eval-01	N	
OEP-F-014	Manual grading	4.4 / Evaluation, Results & Analytics	EvalModule	TC-Eval-02	N	
OEP-F-015	Publish results	4.4 / Evaluation, Results & Analytics	EvalModule	TC-Eval-03	N	
OEP-F-016	Analytics/reports	4.4 / Evaluation, Results & Analytics	EvalModule	TC-Eval-04	N	
OEP-NF-001	Response time	5 / Performance	DeliveryModule	TC-Perf-01	N	
OEP-NF-002	500 concurrent students	5 / Scalability	System	TC-Perf-02	N	
OEP-NF-003	99.5% availability/recovery	5 / Reliability	System	TC-Rel-01	N	
OEP-NF-004	Credential hashing/TLS	5 / Security/Compliance	AuthModule	TC-Sec-01	N	
OEP-NF-005	Responsive WCAG 2.1 AA UI	5 / Usability/Accessibility	UI	TC-UX-01	N	
OEP-SR-001	TLS 1.2+	5.1.2 / Security Requirements	SecurityModule	TC-Sec-02	N	
OEP-SR-002	Password hashing	5.1.2 / Security Requirements	AuthModule	TC-Sec-03	N	
OEP-SR-003	Server-side RBAC	5.1.2 / Security Requirements	AuthModule	TC-Sec-04	N	
OEP-SR-004	Protect exam questions/answers	5.1.2 / Security Requirements	SecurityModule	TC-Sec-05	N	
OEP-SR-005	Audit logging	5.1.2 / Security Requirements	AuditModule	TC-Sec-06	N	